QA | LAB 2

API Testing in Two Layers

Contract is the floor. Behavior is the work. Postman testing where a green suite was still wrong.

What this proves: API testing with Postman and SoapUI that finds the defects contract checks miss, in healthcare and any API driven system.

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Recreated from real healthcare API validation work at CVS Health. No employer data.

The problem

An API passed every contract check and was still wrong in production. The status code was right and the amount was wrong. The field was present and carried the previous member value.

How I built it

1. Contract layer: validated structure, types, required fields, status codes, and documented error shapes. Built a Postman collection, one request per endpoint, no secrets stored in the collection.
2. Behavior layer: tested whether the endpoint enforces its own business rule, or trusts the caller to.
3. Tested duplicate submissions for idempotency. A second identical request must not create a second effect.
4. Tested valid requests for records in the wrong state, and dates outside the plan year.
5. Treated errors as their own surface: shape, message content, and whether an error ever exposed data from another record.
6. Wrote the summary a developer can act on: endpoint, case, expected, actual, evidence.

Validation, expected vs actual

ID	Case	Layer	Expected result
API-01	Valid request, happy path	Both	200, with correct values, not just correct shape
API-02	Missing required field	Contract	400, documented error shape, no internal detail exposed
API-03	Record in wrong state	Behavior	Rule enforced by the endpoint, clear error returned
API-04	Duplicate submission	Behavior	No second effect, duplicate handled as designed
API-05	Date outside plan year	Behavior	Rejected by rule, not by a crash
API-06	Other record data in an error	Security	Error contains no data from another record

Result in this lab: the contract layer was fully green and 2 behavior cases failed. That gap is the point, and it sits at the top of the summary report.

The finding

A green contract suite means the API is eligible to be tested. Nothing more. The defects that matter in healthcare APIs live in behavior: right shape, wrong amount; present field, wrong member. Separating the two layers is what makes those defects findable before production finds them instead.

Deliverables in this lab

- Postman collection structure, described request by request
- 16 API test cases across both layers
- API test summary report sample

Skills shown: API testing, Postman, SoapUI, test cases, defect reporting, healthcare IT, ALM Quality Center, Jira